

- 1.** Load every sheet from the Indian Sales Dataset workbook into separate Pandas DataFrames. Display each sheet name, shape, column names, and the first five rows.
- 2.** For every sheet, inspect column data types and identify columns that should be converted to numeric or datetime types.
- 3.** Create a missing-value report for every sheet showing the count and percentage of missing values in each column.
- 4.** Find duplicate rows in each sheet, count them, display examples, and remove exact duplicates in a separate cleaned DataFrame.
- 5.** Calculate the number of unique values in every column of the Customers, Products, Orders, and Stores sheets. Identify potential key columns.
- 6.** Generate descriptive statistics for numeric columns and identify columns with unusually high or low values for further investigation.
- 7.** Use Boolean indexing to find suspicious records, such as negative quantities, negative sales, impossible customer ages, or discounts outside the expected range.
- 8.** Combine Orders and Order_Items using the appropriate key. Calculate total sales, total cost, total profit, and quantity sold without double-counting orders.
- 9.** Join transaction data with product and category details. Report unmatched product IDs and summarize sales by category and subcategory.
- 10.** Join customer details to order data. Create a customer-level summary with order count, total spending, total profit, average order value, and most recent order date.
- 11.** Use `value_counts()` to report the frequency and percentage distribution of order status, payment mode, sales channel, customer segment, and return reason.
- 12.** Use `groupby()` and `agg()` to create a product-category summary containing total sales, total profit, total quantity, average discount, and distinct order count.
- 13.** Find the top 10 products by total sales, the top 10 by total profit, and the bottom 10 by profit. Use `sorting`, `nlargest()`, or `nsmallest()`.
- 14.** Calculate overall and category-wise profit margins. Handle zero sales safely and avoid division-by-zero errors.
- 15.** Calculate average order value and average quantity per order. Compare these metrics across sales channels, payment modes, and customer segments.
- 16.** Extract year, month, quarter, and weekday from `Order_Date`. Summarize sales and profit by month and quarter.
- 17.** Calculate month-over-month sales growth using `shift()`. Handle the first month and any zero-valued prior-month sales.
- 18.** Calculate average, median, minimum, and maximum shipping and delivery durations using available date columns. Flag invalid date sequences.

- 19.** Analyze returns by month, product, category, and reason. Calculate return counts, returned quantities, refund amounts, and return rates where possible.
- 20.** Find the top 10 states and cities by customer count, sales, and profit. Present the results as sorted DataFrames.
- 21.** Calculate store-wise sales, profit, order count, and profit margin. Merge store area and target data where available, then calculate sales per square foot and target achievement.
- 22.** Compare sales performance by salesperson and sales channel. Rank salespersons by sales, profit, and number of distinct orders.
- 23.** Use `pivot_table()` to create a month-by-category sales summary, and use `crosstab()` to examine the relationship between order status and payment mode.
- 24.** Use `cut()` or `qcut()` to group customers into age bands or spending tiers. Report the number of customers and total sales in each group.
- 25.** Use `transform()` to add category-level sales totals or each order item's share of its order total without changing the original row count.
- 26.** Use `rank()`, `cumsum()`, and cumulative percentage calculations to determine how many products contribute to 80% of total sales.
- 27.** Calculate correlations between available numeric columns such as quantity, discount, sales, cost, and profit. Identify relationships worth investigating further.
- 28.** Use Pandas operations such as `where()`, `select_dtypes()`, `quantile()`, and `replace()` to classify transactions, identify potential outliers, and summarize the results.
- 29.** Create a monthly KPI table containing sales, profit, profit margin, distinct orders, average order value, returned quantity, and target achievement where available.
- 30.** Present 10 evidence-based business insights from the dataset. Support each insight with a numerical result and note assumptions or limitations caused by missing or unmatched data.